

Sales Cloud ↔ ERP — Wiring Diagram

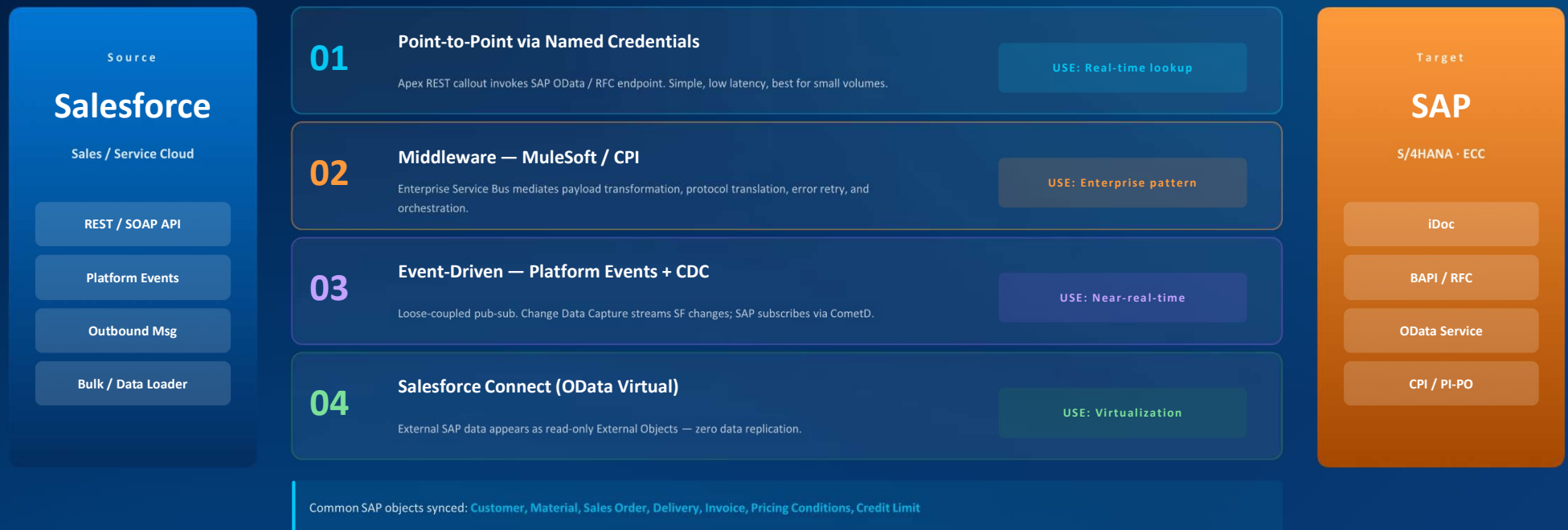
Bi-directional data flows across the Order-to-Cash lifecycle



Common patterns: Outbound Message · Named Credentials + REST callout · Platform Events · Salesforce Connect (OData)

SAP Integration — Connectivity Patterns

Four canonical patterns for wiring Salesforce ↔ SAP



API Catalog & Middleware Concepts

Choosing the right API + orchestration pattern

Salesforce API Catalog

Sync · Low Volume

REST API

JSON · resource-based · <15MB per call · ideal for mobile & web apps

Async · High Volume

Bulk API 2.0

150M records/24hr · CSV · batch load & ETL processes

GraphQL

GraphQL API

Client-driven queries · reduce over-fetching · UX-optimized

Sync · Contract-first

SOAP API

WSDL-based · strict typing · legacy systems & ERP

Event · Pub-Sub

Streaming API

CometD long-polling · Platform Events · CDC

Metadata / Tooling

Metadata API

Deploy/retrieve XML metadata · CI/CD · SFDX

API Governance

Daily request cap by edition · Long-Running Request timeout: 10 min · Composite calls: 25 sub-requests · OAuth 2.0 / JWT auth

Middleware Pipeline · Block Diagram

01

Connector / Adapter

Salesforce / SAP / Oracle / REST connectors — auth & protocol

02

Transformation

Data mapping · schema translation · DataWeave / XSLT / Apex

03

Orchestration

Sequential/parallel flows · conditional branching · aggregation

04

Error Handling & Retry

Dead-letter queue · exponential backoff · idempotency keys

05

Observability & Governance

Monitoring · logging · SLA metrics · API management console

Popular platforms: MuleSoft Anypoint · Dell Boomi · SAP CPI · Informatica · Azure Logic Apps

Data Synchronization & MDM

Golden record strategy across the enterprise data fabric

Sync Pattern Comparison

Real-Time

Push / Callout

Immediate on DML event · REST outbound · low latency · use for critical order events, credit checks

Near-Real-Time

Event Streaming

Pub-Sub · CDC + Platform Events · seconds latency · use for inventory sync, price updates

Batch

Scheduled ETL

Nightly cron · Bulk API · high volume · use for warehouse loads, historical archives

Virtual

Live Federation

Salesforce Connect · External Objects · zero replication · read-only access

Master Data Management · Golden Record Flow

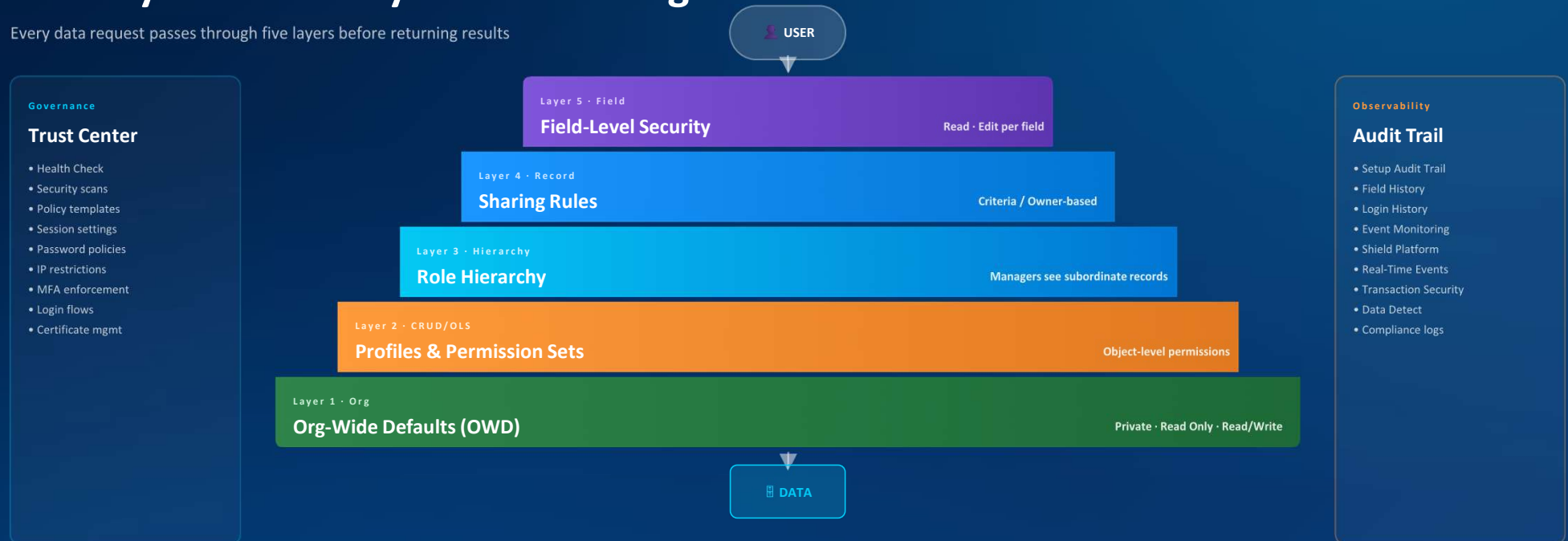


MDM Principles

Single source of truth · Data steward ownership · Survivorship rules · Change lineage tracking · Domains: Customer · Product · Location · Employee

Security Model — Layered Block Diagram

Every data request passes through five layers before returning results



Access Evaluation Order

Salesforce grants the **most permissive** access — checks Profile OR Perm Set for CRUD, then OWD, then widens via Role Hierarchy, Sharing Rules, Manual Sharing, and Team access. Denies at any layer are enforced only where explicitly restricted (e.g. Deny Read on field).

Profiles vs Permission Sets

Baseline access + additive grants — a wiring diagram of user entitlements

One per User (Mandatory)

Profile

Baseline access — the "floor" of a user's permissions

Controls

Object CRUD (Create/Read/Update/Delete)

Field-Level Security defaults

Page Layout assignment

App / Tab visibility · Record Types

Login hours & IP ranges

System permissions (View All, Modify All)

Examples: System Administrator, Standard User, Sales Rep, Read Only

+

Zero or Many per User (Additive)

Permission Set

Extends privileges — grants only, never revokes

Grants

Extra CRUD on specific objects

Additional Field access

Custom permissions & Apex classes

Connected Apps & External Data Sources

Permission Set Groups (bundle multiple)

Muting Permission Sets (exclusions)

Examples: KAM Access, Field Visit Editor, CPQ Admin, Territory Manager

Effective Access

Union of All



1 × Sales Rep Profile

+ Field Visit Editor PS

+ Promotion Mgr PS

+ CRM Analytics Viewer

Result

= Effective Rights ✓

Best practice: minimal Profile + role-based Permission Set Groups → auditable, revocable, scalable

Role Hierarchy & Sharing Rules

Vertical inheritance + horizontal sharing extensions

Block Diagram · Role Hierarchy Tree

Managers inherit access to their team's records



Sharing Rules · Cross-Hierarchy

Grant access laterally, beyond the role tree

Type A

Ownership-Based Sharing

Records owned by Public Group X → shared with Role Y

Type B

Criteria-Based Sharing

Accounts where Region = "West" → shared with West KAM group

Type C

Manual Sharing

Owner grants one-off access via Share button

Type D

Apex-Managed Sharing

Programmatic __Share table insert — complex business logic

Note: Sharing Rules always OPEN access — never restrict.

Data Security, Audit & Governance

Protecting data at rest & in motion — with trail, transparency, and controls

01 · Data Security

Protect the Data

Encryption at Rest

Shield Platform Encryption · AES-256 · deterministic / probabilistic

Encryption in Transit

TLS 1.2+ everywhere · Named Credentials · certificate pinning

Data Classification

Sensitivity tags · Compliance Categorization · PII / PCI / PHI

Data Masking

Sandbox data anonymization · Data Mask managed pkg

IP & Network Access

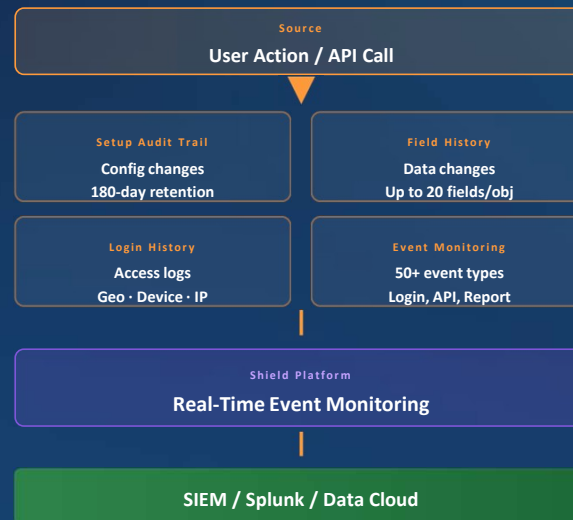
Login IP ranges · Trusted IPs · VPN + Private Connect

MFA & SSO

Salesforce Auth · TOTP · SAML · OAuth · OIDC · Okta / Azure AD

02 · Audit Trail

Track Everything



03 · Governance

Best Practices

Least Privilege Principle

Grant minimum access to perform the job

Segregation of Duties

Separate build, deploy, and audit roles

Quarterly Access Review

SOX / GDPR periodic recertification

Change Management

CI/CD · sandbox promotion · release notes

Health Check Score ≥ 80

Monthly review of security baseline

Backup & Recovery

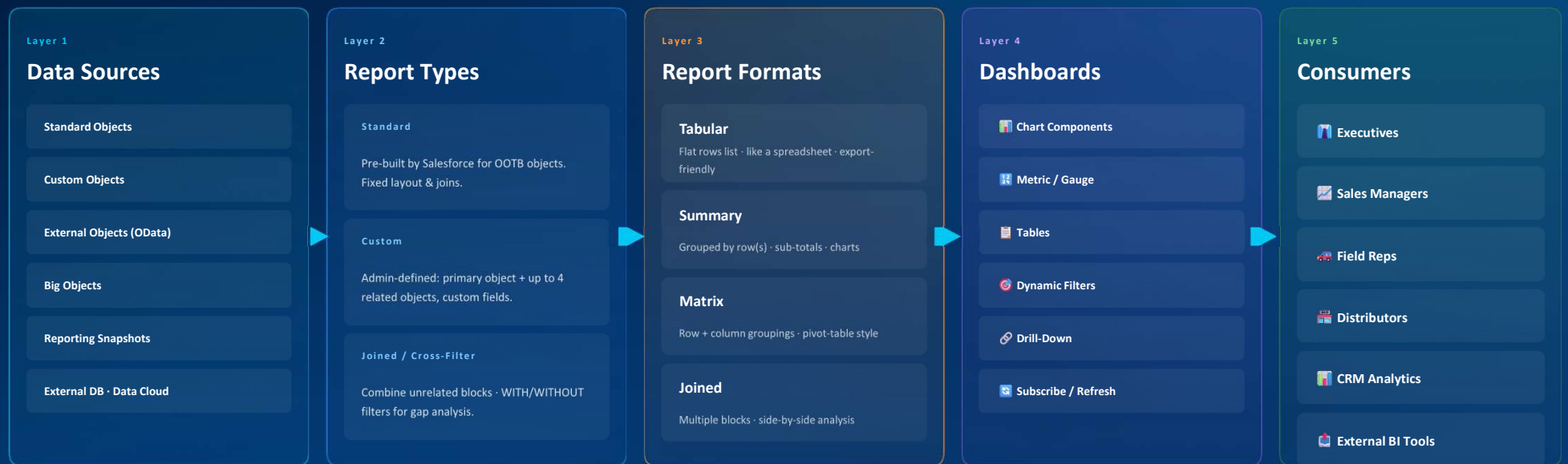
Own Backup / OwnData · daily point-in-time

Documentation & Runbooks

Every security control has an owner + SLA

Reporting Architecture — Block Diagram

From raw records to insight — the Salesforce analytics stack



Report & Dashboard folders enforce sharing · Access follows the same OWD + sharing rule model